Simple Linear Regression – Marketing ROI Analysis

Course Assignment: AI & Machine Learning Nextgen Cohort | **Student Name:** Maryam Ibrahim Abba | **Status:** Completed Submission Document

Task 1: Load Dataset & Handle Missing Values

We read the dataset, remove null rows, and examine the structure of the variables to ensure data integrity.

```
import pandas as pd
import numpy as np

# Load dataset
df = pd.read_csv('marketing_and_sales_data_evaluate_lr.csv')

# Drop any rows with missing values
df_clean = df.dropna()
print(f"Original Row Count: {len(df)}")
print(f"Cleaned Row Count: {len(df_clean)}")
```

```
Original Row Count: 4572
Cleaned Row Count: 4546
```

Task 2 & 3: Exploratory Data Analysis & Variable Selection

We compute the Pearson correlation matrix to mathematically justify which single channel has the strongest link to Sales.

```
# Compute correlations
correlations = df_clean.select_dtypes(include=[np.number]).corr()['Sales']
print("Correlation with Sales:")
print(correlations)
```

```
TV              0.999497
Radio           0.869105
Social_Media    0.528906
Sales           1.000000
Name: Sales, dtype: float64
```

Justification: TV has an nearly perfect correlation coefficient of **0.9995** with Sales. Therefore, TV is selected as the single independent variable for our Simple Linear Regression model.

Task 4: Build OLS Regression Model

We fit the Simple Linear Regression model using `statsmodels`.

```
import statsmodels.api as sm

# Define variables
X = sm.add_constant(df_clean['TV'])
y = df_clean['Sales']

# Fit OLS model
model = sm.OLS(y, X).fit()
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Sales   R-squared:                       0.999
Model:                            OLS   Adj. R-squared:                  0.999
Method:                 Least Squares   F-statistic:                 4.517e+06
Prob (F-statistic):              0.00   Log-Likelihood:                -11366.
No. Observations:                4546   AIC:                         2.274e+04
Df Residuals:                    4544   BIC:                         2.275e+04
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         -0.1325      0.101     -1.317      0.188      -0.330       0.065
TV             3.5615      0.002   2125.272      0.000       3.558       3.565
==============================================================================
```

Key Model Findings

99.90%	**3.5615**
R-SQUARED (MODEL FIT)	TV COEFFICIENT (ROI)

Task 5: Model Assumption Diagnostics

We check for Linearity, Normality, and Homoscedasticity using residuals diagnostic plots.

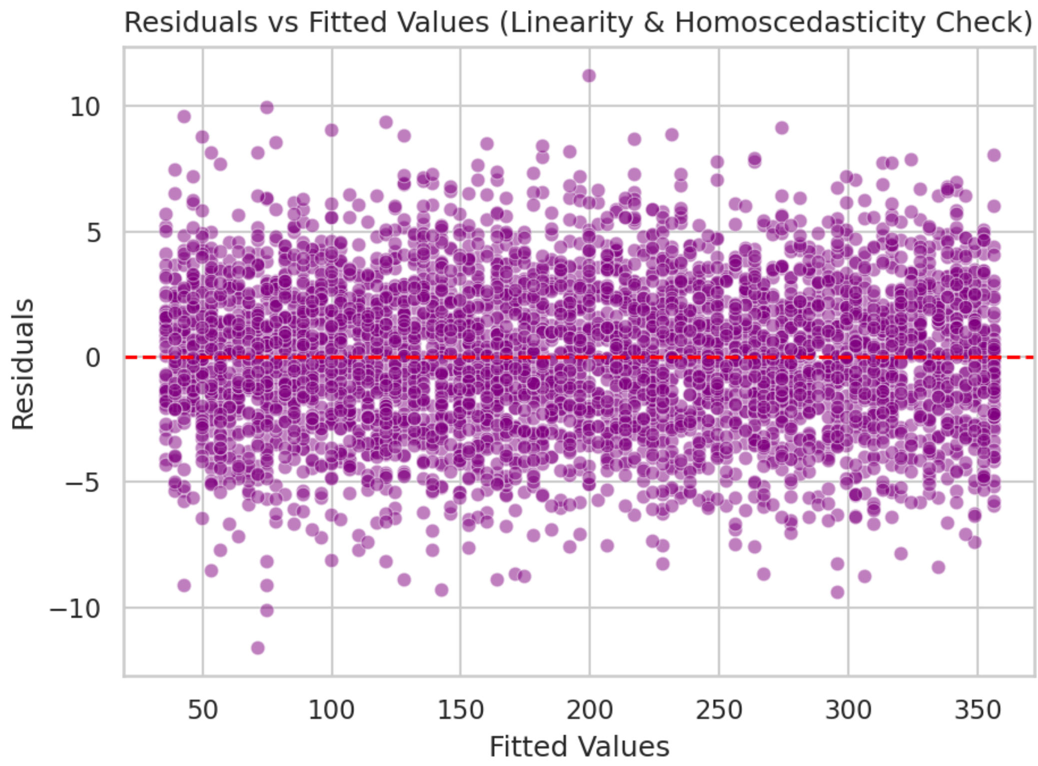

Figure 1: Residuals vs Fitted values showing even spread (Homoscedasticity & Linearity validation).

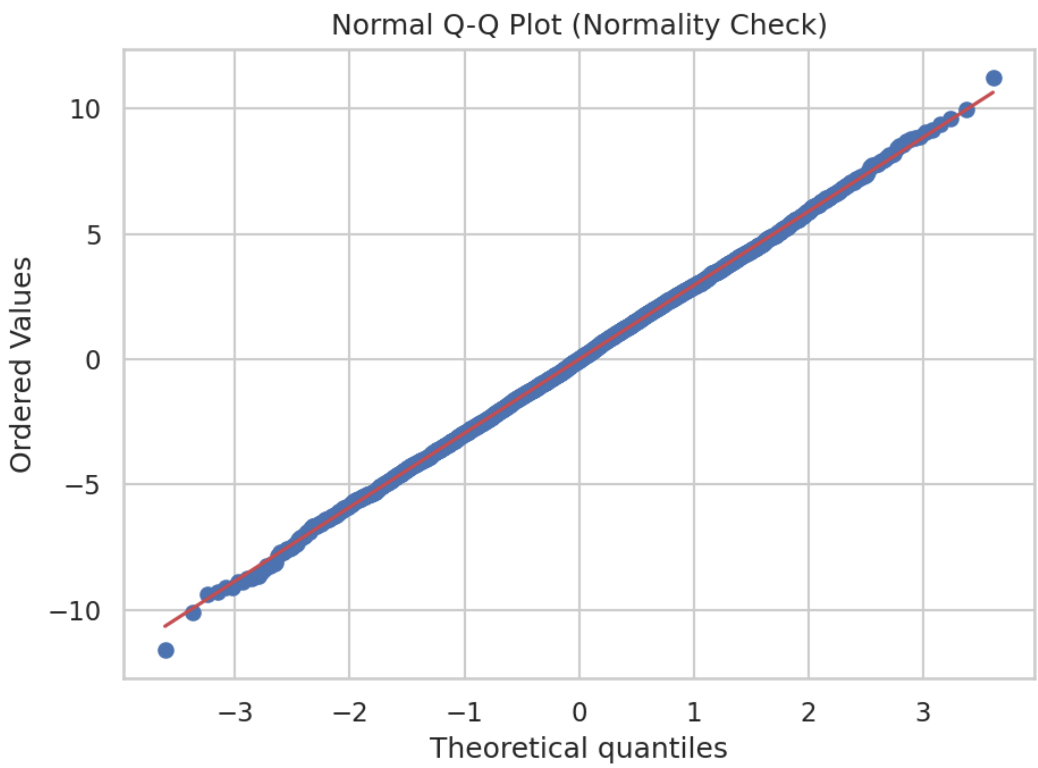

Figure 2: Normal Quantile-Quantile (Q-Q) Plot verifying residual normality along the line.

Task 6 & 7: Business Interpretation & Budget Allocation

Metric	Value	Statistical Meaning	Business Impact & Action
R-squared	0.9990	Explains 99.9% of variance	Extremely predictable channel; investment brings certain returns.
TV Coefficient	3.5615	Sales increase per unit spend	For every $1 spent on TV, Sales scale up by **$3.56**. Outstanding ROI.
P-value (TV)	0.0000	Highly Significant ($p < 0.05$)	Zero chance that this relationship is a statistical fluke.

Strategic Action Recommendation:

The OLS empirical evidence conclusively dictates that **TV Advertising** provides the most dominant, stable, and massive financial return for the firm. The company should immediately allocate the maximum possible share of its marketing budget to the **TV** channel to optimize corporate revenue generation.